Unit 1 - Quantum Mechanics

Quantum mechanics is a fundamental theory in physics that describes the physical properties of nature at the scale of atoms and subatomic particles. It is the foundation of all quantum physics including quantum chemistry, quantum field theory, quantum technology, and quantum information science.

Some key concepts in quantum mechanics include: - Wave-particle duality: the concept that every elementary particle or quantic entity exhibits the properties of not only particles, but also waves. - Uncertainty principle: the principle that the position and momentum of a particle cannot both be precisely determined at the same time. - Superposition: the ability of a quantum system to be in multiple states at the same time. - Entanglement: a phenomenon where two or more particles become interconnected and the state of each particle cannot be described independently of the others.

Quantum mechanics has been extremely successful in explaining a wide range of phenomena, including the behavior of atoms and molecules, the properties of materials, and the fundamental interactions between particles. It has also led to the development of many technologies, such as transistors, lasers, and magnetic resonance imaging (MRI).

Inadequacy of classical mechanics for the notes of the Unit 1 - Quantum Mechanics in the subject of ENGINEERING PHYSICS

1. Classical mechanics, also known as Newtonian mechanics, is a branch of physics that deals with the motion of macroscopic objects based on the three laws of motion formulated by Sir Isaac Newton.
2. Classical mechanics is highly successful in explaining the motion of macroscopic objects, but it fails to explain the behavior of microscopic particles such as atoms and subatomic particles.
3. Some phenomena that cannot be explained by classical mechanics include the photoelectric effect, blackbody radiation, and the stability of atoms.
4. The photoelectric effect is the emission of electrons from a metal surface when light shines on it. According to classical mechanics, the energy of the emitted electrons should increase with the intensity of the light, but experiments showed that the energy of the emitted electrons depends only on the frequency of the light and not on its intensity.
5. Blackbody radiation is the electromagnetic radiation emitted by a body in thermal equilibrium. Classical mechanics predicts that the energy emitted by a blackbody at high frequencies should be infinite, but this is not observed in experiments. This problem is known as the ultraviolet catastrophe.
6. According to classical mechanics, electrons orbiting the nucleus of an atom should continuously lose energy and eventually spiral into the nucleus, but this is not observed in nature. Atoms are stable and electrons do not spiral into the nucleus.

7. These phenomena and others led to the development of quantum mechanics, a new branch of physics that successfully explains the behavior of microscopic particles.

Planck's theory of black body radiation (qualitative)

- Planck's radiation law is a mathematical relationship formulated in 1900 by German physicist Max Planck to explain the spectral-energy distribution of radiation emitted by a blackbody.
- A blackbody is a hypothetical body that completely absorbs all radiant energy falling upon it, reaches some equilibrium temperature, and then reemits that energy as quickly as it absorbs it.
- Planck's law describes the spectral density of electromagnetic radiation emitted by a black body in thermal equilibrium at a given temperature T, when there is no net flow of matter or energy between the body and its environment.
- Using Planck's law, a black body's emitted radiation may be termed by its frequency variation. Planck's radiation, a form of thermal radiation, is another name for this type of radiation. The greater the temperature of the body, the more radiation of all wavelengths is emitted by it.
- The blackbody radiation problem was solved in 1900 by Max Planck. Planck used the same idea as the Rayleigh–Jeans model in the sense that he treated the electromagnetic waves between the walls inside the cavity classically, and assumed that the radiation is in equilibrium with the cavity walls.

Compton Effect

The Compton effect refers to the increase in the wavelength of photons (X-rays or gamma rays), due to their scattering by a charged particle (usually an electron). The impact has ended up being one of the foundations of quantum mechanics, which represents both wave and particle properties of radiation .

Quantum Mechanics has been an answer to observations that classical physics couldn't explain. The Compton Effect is one such phenomenon. The Compton Effect was a breakthrough in the 20th century, as it confirmed the particle nature of radiation .

By classical theory, when an electromagnetic wave is scattered off atoms, the wavelength of the scattered radiation is expected to be the same as the wavelength of the incident radiation. However, the Compton effect is the term used for an unusual result observed when X-rays are scattered on some materials .

The Compton effect is also called Compton scattering and is a principal way in which radiant energy is absorbed in matter .

de-Broglie concept of matter waves

- According to the de Broglie concept of matter waves, the matter has dual nature. It means when the matter is moving it shows the wave properties (like interference, diffraction etc.) are associated with it and when it is in the state of rest then it shows particle properties. Thus the matter has dual nature.
- Using the concept of the electron matter wave, de Broglie provided a rationale for the quantization of the electron's angular momentum in the hydrogen atom, which was postulated in Bohr's quantum theory.
- Compton's formula established that an electromagnetic wave can behave like a particle of light when interacting with matter. In 1924, Louis de Broglie proposed a new speculative hypothesis that electrons and other particles of matter can behave like waves. Today, this idea is known as de Broglie's hypothesis of matter waves.
- The De Broglie hypothesis proposes that all matter exhibits wave-like properties and relates the observed wavelength of matter to its momentum. After Albert Einstein's photon theory became accepted, the question became whether this was true only for light or whether material objects also exhibited wave-like behavior.
- De Broglie, in his 1924 PhD thesis, proposed that just as light has both wave-like and particle-like properties, electrons also have wave-like properties.

Davisson and Germer Experiment

The Davisson-Germer experiment was conducted in 1927 by two scientists, C.J. Davisson and L.H. Germer, at Bell Labs. The experiment demonstrated the wave nature of the electron, confirming the earlier hypothesis of deBroglie . Putting wave-particle duality on a firm experimental footing, it represented a major step forward in the development of quantum mechanics.

In the experiment, slow-moving electrons were fired at a crystalline nickel target. The angular dependence of the reflected electron intensity was measured and was determined to have the same diffraction pattern as those predicted by Bragg for X-rays.

The basic thought behind the Davisson and Germer experiment was that the waves reflected from two different atomic layers of a Ni crystal will have a fixed phase difference. After reflection, these waves will interfere either constructively or destructively, hence producing a diffraction pattern.

This experiment, for the first time, proved the wave nature of electrons and verified the de Broglie equation. de Broglie argued the dual nature of matter back in 1924, but it was only later that Davisson and Germer experiment verified the results. The results established the first experimental proof of quantum mechanics.

Phase velocity and group velocity

Phase and group velocity are two important and related concepts in wave mechanics. They arise in quantum mechanics in the time development of the state function for the continuous case, i.e. wave packets.

- The phase velocity of a wave is the rate at which the wave propagates in any medium. This is the velocity at which the phase of any one frequency component of the wave travels.
- In an anomalous medium, the group velocity is greater than the phase velocity. In a normal medium, the phase velocity is greater than the group velocity.
- The group velocity is positive, while the phase velocity is negative.

These concepts are important in understanding the speed of a particle in quantum mechanics. The relationship between phase velocity and group velocity can be shown using the de Broglie postulates.

Time-dependent and time-independent Schrodinger wave equations

The Schrödinger equation is a mathematical equation that describes the evolution of a quantum mechanical system over time. It is named after the physicist Erwin Schrödinger, who derived the equation in 1925. The Schrödinger equation is a partial differential equation that describes how the wavefunction of a physical system changes over time.

There are two forms of the Schrödinger equation: the time-dependent Schrödinger equation and the time-independent Schrödinger equation.

1. **Time-dependent Schrödinger equation:** This form of the equation is used to describe the evolution of a quantum system over time. It is given by the equation:

   `iħ  / t = H`

   where `i` is the imaginary unit, `ħ` is the reduced Planck constant, is the wavefunction of the system, `t` is time, and `H` is the Hamiltonian operator, which represents the total energy of the system.

2. **Time-independent Schrödinger equation:** This form of the equation is used to describe the stationary states of a quantum system, where the wavefunction does not change over time. It is given by the equation:

   `H  = E`

   where `H` is the Hamiltonian operator, is the wavefunction of the system, and `E` is the total energy of the system.

The time-independent Schrödinger equation is often used to solve for the stationary states of a system, while the time-dependent Schrödinger equation is used to describe how the system evolves over time.

These equations are fundamental to the study of quantum mechanics and are used to describe a wide range of physical phenomena, including the behavior of particles in a potential well, the energy levels of atoms and molecules, and the scattering of particles. They are also used to model the behavior of macroscopic systems, such as the flow of electrons in a semiconductor or the behavior of a superconductor.

Physical interpretation of wave function

- The wave function, at a particular time, contains all the information that anybody at that time can have about the particle.
- The wave function itself has no physical interpretation. It is not measurable.
- However, the square of the absolute value of the wave function has a physical interpretation.
- There is no physical meaning of wave function as it is not a quantity which can be observed. Instead, it is complex.
- The standard assumption is that the wave function of an electron is a probability amplitude, and its modulus square gives the probability density of finding the electron in a certain location at a given instant.

Particle in a One-Dimensional Box

The particle in a one-dimensional box is a fundamental problem in quantum mechanics that demonstrates the quantization of energy levels. It is a simple model that describes the behavior of a particle confined to a one-dimensional region of space.

1. The potential energy of the particle is zero inside the box and infinite outside the box. This means that the particle is confined to the box and cannot escape.
2. The wave function of the particle must be zero at the boundaries of the box, as the probability of finding the particle outside the box is zero.
3. The Schrödinger equation can be solved for this system to obtain the allowed energy levels and wave functions of the particle.
4. The energy levels are quantized, meaning that the particle can only have certain discrete energy values.
5. The wave functions of the particle are sinusoidal and represent the probability density of finding the particle at a particular position within the box.
6. The particle in a one-dimensional box is a useful model for understanding the behavior of electrons in quantum wells and other confined systems.

Unit 2 - Electromagnetic Field Theory

Electromagnetic Field Theory is the study of electromagnetic fields and their interactions with charged particles. It is a fundamental concept in physics and is used to explain a wide range of phenomena, including the behavior of light, electricity, and magnetism.

Some key concepts in Electromagnetic Field Theory include:

1. **Electric fields** are produced by electric charges and can exert forces on other charges.
2. **Magnetic fields** are produced by moving electric charges and can exert forces on other moving charges.
3. **Electromagnetic waves** are disturbances in electric and magnetic fields that can travel through space.
4. **Maxwell's equations** describe how electric and magnetic fields are generated and how they interact with each other and with charges.
5. **Electromagnetic induction** is the process by which a changing magnetic field can induce an electric current in a conductor.

Electromagnetic Field Theory has numerous applications, including the design of electric motors, generators, and transformers, as well as the transmission and reception of radio and television signals.

Basic concept of Stoke's theorem and Divergence theorem for the notes of the Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS

Stoke's Theorem

- Stoke's theorem is a statement about the integration of differential forms on manifolds.
- It relates the integral of a differential form over the boundary of some orientable manifold to the integral of its exterior derivative over the whole manifold.
- Mathematically, it is expressed as: M = Md , where M is the boundary of the manifold M, is a differential form, and d is its exterior derivative.

Divergence Theorem

- The divergence theorem, also known as Gauss's theorem, is a result that relates the flow of a vector field through a closed surface to the behavior of the vector field inside the surface.
- It states that the outward flux of a vector field through a closed surface is equal to the volume integral of the divergence of the field within the surface.
- Mathematically, it is expressed as: V**F** · d**A** = Vdiv **F** dV, where V is the boundary of the volume V, **F** is a vector field, and div **F** is its divergence.

These theorems are important concepts in the study of Electromagnetic Field Theory and are commonly used in the analysis of electric and magnetic fields. They provide a way to relate the behavior of a field within a region to its behavior on the boundary of the region.

Unit 2 - Electromagnetic Field Theory

Basic laws of electricity and magnetism

1. **Like electric charges repel, and unlike electric charges attract.** The force of attraction or repulsion is inversely proportional to the square of the distance between them .
2. **Magnetic poles always exist as north-south pairs** .
3. **An electric current in a wire generates a magnetic field around the wire** .
4. **Moving a loop of wire toward or away from a magnetic field induces a current in the wire** .
5. **Electricity can exist without magnetism, but magnetism cannot exist without electricity** .
6. The operation of electric motors is governed by various laws of electricity and magnetism, including **Faraday's law of induction, Ampère's circuital law, Lenz' law, and the Lorentz force** .
7. **Lorentz force law** states that when a magnetic field is created, a magnetic force acts on any object that lies inside that magnetic field. The magnitude of this magnetic force is given by Lorentz force law, which takes into account both the magnetic and electric force around an object .
8. The four basic laws of electricity and magnetism had been discovered experimentally through the work of physicists such as **Oersted, Coulomb, Gauss, and Faraday**. Maxwell discovered logical inconsistencies in these earlier results and identified the incompleteness of Ampère's law as their cause .

Continuity equation for current density for the notes of the Unit 2 - Electromagnetic Field Theory in the subject of ENGINEERING PHYSICS

The continuity equation for current density is a mathematical expression that describes the conservation of electric charge. It states that the rate of change of electric charge in any volume of space is equal to the current flowing into the volume minus the current flowing out of the volume.

Mathematically, the continuity equation for current density is expressed as:

- **J** = - / t

where **J** is the current density vector, is the charge density, and t is time.

The continuity equation for current density can be derived from the principle of conservation of electric charge, which states that electric charge can neither be created nor destroyed. This principle implies that the total amount of charge in any closed system must remain constant.

In the context of electromagnetic field theory, the continuity equation for current density is an important tool for analyzing the behavior of electric currents and charges in various situations. It is commonly used in the study of electric circuits, electromagnetic waves, and other phenomena involving the flow of electric charge.

Some key points to remember about the continuity equation for current density are:

- It describes the conservation of electric charge.
- It relates the rate of change of charge density to the current density.
- It is derived from the principle of conservation of electric charge.
- It is an important tool in the study of electromagnetic field theory.

Displacement Current

- Displacement current is a term used in electromagnetism and is defined in terms of the rate of change of the electric displacement field .
- It has the same units as electric current density and is a source of the magnetic field just as actual current is .
- According to Faraday's law of electromagnetic induction, a time-varying magnetic field induces an emf. According to Maxwell, an electric field sets up a current and hence a magnetic field. Such a current is called displacement current .
- A time-varying electric field produces a magnetic field and vice-versa .
- Displacement current plays a major role during the propagation of electromagnetic radiations like light waves and radio waves through space .

Unit 2 - Electromagnetic Field Theory: Maxwell's Equations in Integral and Differential Form

Maxwell's equations are a set of four partial differential equations that describe the behavior of the electric and magnetic fields. These equations were first presented by James Clerk Maxwell in the 19th century. They are fundamental to the study of electromagnetic field theory and have numerous applications in physics and engineering.

The four Maxwell's equations in integral form are:

1. Gauss's Law: This law states that the electric flux through any closed surface is equal to the charge enclosed within the surface divided by the permittivity of free space. Mathematically, it is represented as:
$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

2. Gauss's Law for Magnetism: This law states that the magnetic flux through any closed surface is zero. Mathematically, it is represented as:

$$\oint \vec{B} \cdot d\vec{A} = 0$$

3. Faraday's Law of Induction: This law states that the electromotive force (EMF) induced in a closed loop is equal to the negative rate of change of the magnetic flux through the loop. Mathematically, it is represented as:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

4. Ampere's Law with Maxwell's Correction: This law states that the magnetic field around a closed loop is equal to the sum of the current flowing through the loop and the displacement current, which is the rate of change of the electric flux through the loop. Mathematically, it is represented as:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(I + \epsilon_0 \frac{d\Phi_E}{dt} \right)$$

The four Maxwell's equations can also be expressed in differential form. The differential form of the equations is more commonly used in theoretical and computational work. The differential form of the equations is obtained by applying the divergence and curl operators to the electric and magnetic fields.

The four Maxwell's equations in differential form are:

1. Gauss's Law:

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

2. Gauss's Law for Magnetism:

$$\nabla \cdot \vec{B} = 0$$

3. Faraday's Law of Induction:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

4. Ampere's Law with Maxwell's Correction:

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

These equations provide a complete description of the behavior of the electric and magnetic fields and are essential for the study of electromagnetic field theory. They have numerous applications in physics and engineering, including the design of electric motors, generators, transformers, and antennas.

Unit 2 - Electromagnetic Field Theory

Maxwell's Equations in Vacuum and in Conducting Medium

Maxwell's equations are a set of four partial differential equations that describe the behavior of the electric and magnetic fields. These equations were first presented by James Clerk Maxwell in the 19th century. The equations are as follows:

1. Gauss's Law: This law states that the electric flux through any closed surface is equal to the charge enclosed within the surface divided by the permittivity of free space. Mathematically, it is represented as: • **E** = / 0
2. Gauss's Law for Magnetism: This law states that the magnetic flux through any closed surface is zero. Mathematically, it is represented as: • **B** = 0
3. Faraday's Law of Induction: This law states that the electromotive force (EMF) induced in a closed loop is equal to the negative of the rate of change of the magnetic flux through the loop. Mathematically, it is represented as: x **E** = - **B**/ t
4. Ampere's Law with Maxwell's Correction: This law states that the magnetic field around a closed loop is equal to the sum of the current flowing through the loop and the displacement current, which is the rate of change of the electric flux through the loop. Mathematically, it is represented as: x **B** = 0(**J** + 0 **E**/ t)

In a conducting medium, the electric field **E** is related to the current density **J** by the conductivity of the medium, as **J** = **E**. This relationship can be used to modify Maxwell's equations in a conducting medium.

Poynting vector and Poynting theorem

- The Poynting vector is used throughout electromagnetics in conjunction with Poynting's theorem, the continuity equation expressing conservation of electromagnetic energy, to calculate the power flow in electric and magnetic fields.
- Poynting's theorem states that the cross product of electric field vector, E and magnetic field vector, H at any point is a measure of the rate of flow of electromagnetic energy per unit area at that point, that is P = E x H.
- The Poynting vector is named after its discoverer, J.H. Poynting.
- In physics, the Poynting vector represents the directional energy flux (the energy transfer per unit area per unit time) or power flow of an electromagnetic field.
- The SI unit of the Poynting vector is the watt per square metre (W/m 2); kg/s 3 in base SI units.
- Poynting's vector is a measure of the momentum density carried by the electromagnetic field.

- In electrodynamics, Poynting's theorem is a statement of conservation of energy for electromagnetic fields developed by British physicist John Henry Poynting.
- It states that in a given volume, the stored energy changes at a rate given by the work done on the charges within the volume, minus the rate at which energy leaves the volume.

Plane Electromagnetic Waves in Vacuum and Their Transverse Nature

Unit 2 - Electromagnetic Field Theory in the Subject of Engineering Physics

1. Electromagnetic waves are transverse waves that consist of oscillating electric and magnetic fields that propagate through space.
2. In a vacuum, these waves travel at the speed of light, which is approximately 299,792,458 meters per second.
3. The electric and magnetic fields of an electromagnetic wave are perpendicular to each other and to the direction of propagation of the wave.
4. This means that the electric and magnetic fields oscillate in planes that are perpendicular to the direction in which the wave is traveling.
5. The transverse nature of electromagnetic waves can be demonstrated by the polarization of light. Polarization refers to the orientation of the electric field of an electromagnetic wave.
6. When light passes through a polarizing filter, only the component of the electric field that is parallel to the filter's transmission axis is transmitted, while the component that is perpendicular to the transmission axis is absorbed.
7. This results in the light being polarized, with its electric field oscillating in a single plane.
8. The transverse nature of electromagnetic waves has important implications for the behavior of these waves when they interact with matter, such as reflection, refraction, and diffraction.

Relation between electric and magnetic fields of an electromagnetic wave

1. Electromagnetic waves are transverse waves that consist of oscillating electric and magnetic fields that are perpendicular to each other and to the direction of propagation of the wave.
2. The electric and magnetic fields of an electromagnetic wave are in phase and their magnitudes are related by the equation `E = cB`, where `E` is the electric field strength, `B` is the magnetic field strength, and `c` is the speed of light.
3. The electric and magnetic fields of an electromagnetic wave are mutually dependent and self-sustaining. A changing electric field generates a magnetic field, and a changing magnetic field generates an electric field.

4. The energy of an electromagnetic wave is equally divided between its electric and magnetic fields.
5. The Poynting vector, which represents the direction and magnitude of the energy flow of an electromagnetic wave, is given by the cross product of the electric and magnetic fields: `S = E x H`, where `S` is the Poynting vector, `E` is the electric field, and `H` is the magnetic field.
6. The electric and magnetic fields of an electromagnetic wave obey the wave equation, which describes how the fields change over time and space. The wave equation for the electric field is given by $^2E = \quad ^2E/ t^2$, where ^{2} is the Laplace operator, is the permeability of free space, is the permittivity of free space, and $^2E/ t^2$ is the second time derivative of the electric field. The wave equation for the magnetic field is similar, with `E` replaced by `B`.
7. The electric and magnetic fields of an electromagnetic wave are orthogonal to each other and to the direction of propagation of the wave. This means that the electric field, magnetic field, and direction of propagation of the wave are all perpendicular to each other.
8. The electric and magnetic fields of an electromagnetic wave are transverse to the direction of propagation of the wave. This means that the fields oscillate in a plane that is perpendicular to the direction of propagation of the wave.

Plane Electromagnetic Waves in Conducting Medium

Unit 2 - Electromagnetic Field Theory

Subject: ENGINEERING PHYSICS

1. A plane electromagnetic wave is a wave that travels through a conducting medium.
2. The wave consists of an electric field and a magnetic field that oscillate perpendicular to each other and to the direction of wave propagation.
3. The electric field induces a current in the conducting medium, which in turn generates a magnetic field.
4. The magnetic field induces an electric field, and the process repeats, allowing the wave to propagate through the medium.
5. The presence of the conducting medium causes the wave to attenuate, or decrease in amplitude, as it travels.
6. The rate of attenuation depends on the conductivity of the medium and the frequency of the wave.
7. The wave can also be reflected or refracted at the interface between two different conducting media.
8. The behavior of plane electromagnetic waves in conducting media is described by Maxwell's equations.

Skin Depth

Skin depth is a measure of the distance over which an electromagnetic wave entering a conducting surface is damped and reduces in amplitude by a factor of 1/e . It is given by the formula `1/√f` where `f` is the coil frequency, is the magnetic permeability of the material, and is the electrical conductivity of the material .

- Skin depth is the distance over which the magnitude of the electric or magnetic field is reduced by a factor of 1/e .
- Since power is proportional to the square of the field magnitude, skin depth may also be interpreted as the distance at which the power in the wave is reduced by a factor of (1/e)^2 .
- Skin depth depends on the frequency of the current and the electrical and magnetic properties of the conductor .

This concept is important in electromagnetic field theory and has applications in areas such as induction cooking, where stranded coils are used to reduce heating of the coil itself due to skin effect .

Unit 3 - Wave Optics

Wave optics, also known as physical optics, is the branch of optics that studies the behavior of light as a wave. It is concerned with the phenomena of interference, diffraction, and polarization of light.

1. **Interference:** Interference is the phenomenon in which two or more waves superimpose to form a resultant wave of greater, lower, or the same amplitude. Constructive interference occurs when the phase difference between the waves is a multiple of 2 , whereas destructive interference occurs when the phase difference is an odd multiple of .
2. **Diffraction:** Diffraction is the bending of waves around obstacles or the spreading out of waves when they pass through an aperture. The amount of diffraction depends on the wavelength of the wave and the size of the obstacle or aperture.
3. **Polarization:** Polarization is the property of electromagnetic waves, such as light, that describes the direction of the electric field. Unpolarized light consists of waves with electric fields in all directions, whereas polarized light consists of waves with electric fields in a specific direction.

These phenomena can be explained using the principle of superposition and Huygens' principle. The principle of superposition states that the total displacement of a wave at any point is the sum of the individual displacements of all the waves present at that point. Huygens' principle states that every point on a wavefront can be considered as a source of secondary wavelets, which spread out in all directions with the same speed as the wave.

Wave optics has many practical applications, including the design of optical instruments, such as microscopes and telescopes, and the development of technologies, such as fiber optics and holography. It is also used to study the behavior of light in various media, such as air, water, and glass.

Coherent sources for the notes of the Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS

- Coherent sources are sources of light that emit waves of the same frequency and constant phase difference.
- The waves from coherent sources interfere constructively and destructively, producing an interference pattern.
- Coherent sources can be produced by dividing a single light source into two or more beams using a beam splitter or by using lasers.
- Lasers produce highly coherent light, making them ideal for applications such as holography and interferometry.
- Coherence is a measure of the correlation between two waves and is important in many optical phenomena, including interference, diffraction, and polarization.
- Coherent sources are essential for many applications in science, engineering, and technology, including optical communication, metrology, and medical imaging.

Interference in Uniform and Wedge Shaped Thin Films

Interference is a phenomenon in which two or more waves superimpose to form a resultant wave of greater, lower, or the same amplitude. Interference usually refers to the interaction of waves that are correlated or coherent with each other, either because they come from the same source or because they have the same or nearly the same frequency.

In the context of uniform and wedge shaped thin films, interference occurs when light waves reflect off the two surfaces of the film and interfere with each other. The interference pattern depends on the thickness of the film, the angle of incidence of the light, and the refractive index of the film.

1. **Uniform Thin Films:** In a uniform thin film, the thickness of the film is constant. When light is incident on the film, some of it is reflected at the first surface, while the rest is transmitted and then reflected at the second surface. The two reflected waves then interfere with each other, producing an interference pattern. The condition for constructive interference is given by $2nt = (m + 1/2)$, where n is the refractive index of the film, t is the thickness of the film, m is an integer, and is the wavelength of the light in vacuum.

2. **Wedge Shaped Thin Films:** In a wedge shaped thin film, the thickness of the film varies linearly. When light is incident on the film, it is reflected at both surfaces, producing an interference pattern. The condition for constructive interference is given by $2nt = m$, where n is the refractive

index of the film, t is the thickness of the film at the point of reflection, m is an integer, and is the wavelength of the light in vacuum.

These are the basic concepts of interference in uniform and wedge shaped thin films. It is important to understand these concepts in order to study the behavior of light in thin films and its applications in various fields such as optics and engineering.

Necessity of extended sources for the notes of the Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS

1. **Extended sources** are important for understanding the behavior of light waves in wave optics.
2. They provide a more **realistic** representation of light sources, as most light sources in nature are extended rather than point sources.
3. The use of extended sources allows for the study of **interference** and **diffraction** phenomena, which are crucial concepts in wave optics.
4. The study of extended sources also enables the understanding of the **spatial coherence** of light waves, which is important for applications such as holography and interferometry.
5. In summary, the inclusion of extended sources in the notes for Unit 3 - Wave Optics is necessary for a **comprehensive** understanding of the subject matter.

Newton's Rings and its applications

Newton's rings are a phenomenon in which an interference pattern is created by the reflection of light between two surfaces, typically a spherical surface and an adjacent touching flat surface. It is named after Isaac Newton, who investigated the effect in 1666.

Some of the applications of Newton's rings are as follows:

1. For testing the uniformity of a polished surface by studying the interference pattern the surface makes when placed in contact with a perfectly flat glass surface.
2. To verify the thickness of a surface.
3. To test out whether a surface is uniform.
4. To measure the thickness of thin films.
5. To measure the shape of lenses and other curved surfaces.
6. To measure the refractive index of materials.

Introduction to Diffraction

Diffraction is the phenomenon of bending of light around the corners of an obstacle or aperture into the region of geometrical shadow of the obstacle. It is a characteristic property of waves, including light waves, sound waves, and water waves. Diffraction is observed when the size of the obstacle or aperture is comparable to the wavelength of the wave.

Some key points to remember about diffraction are:

1. Diffraction is a wave property and is observed for all types of waves.
2. The amount of diffraction depends on the size of the obstacle or aperture relative to the wavelength of the wave.
3. Diffraction is more pronounced when the wavelength of the wave is larger compared to the size of the obstacle or aperture.
4. Diffraction can cause the spreading out of light as it passes through a narrow slit or around a sharp edge.
5. The diffraction pattern produced by a single slit consists of a central bright fringe, with alternating dark and bright fringes on either side.

Fraunhoffer Diffraction at Single Slit and Double Slit

Unit 3 - Wave Optics in the subject of ENGINEERING PHYSICS

1. Fraunhoffer diffraction is a type of diffraction that occurs when a wave passes through an aperture or around an obstacle and the diffracted wavefronts are essentially parallel.
2. This type of diffraction is also known as far-field diffraction.
3. Fraunhoffer diffraction can be observed at a single slit or a double slit.
4. In the case of a single slit, the diffraction pattern is characterized by a central bright fringe, flanked by a series of dark and bright fringes of decreasing intensity.
5. The width of the central bright fringe is inversely proportional to the width of the slit.
6. In the case of a double slit, the diffraction pattern is characterized by a series of bright and dark fringes, with the bright fringes being equally spaced.
7. The spacing between the bright fringes is inversely proportional to the distance between the two slits.
8. The intensity of the bright fringes in a double slit diffraction pattern is also affected by the width of the slits and the distance between them.
9. Fraunhoffer diffraction is an important concept in the study of wave optics and has numerous applications in fields such as engineering and physics.

Absent Spectra

Absent spectra is a topic in the Wave Optics module of Engineering Physics. It is related to the study of Fraunhoffer diffraction at a single slit and at a double slit . It is also related to the study of diffraction grating, spectra with grating, and dispersive power of grating .

1. **Fraunhoffer diffraction** is the study of the diffraction of light when the distance between the source and the screen is much larger than the size of

the diffracting aperture.

2. **Diffraction grating** is an optical component with a periodic structure that splits and diffracts light into several beams traveling in different directions.
3. **Dispersive power of grating** is the ability of a diffraction grating to separate light of different wavelengths.

Diffraction Grating

- A diffraction grating is an optical component with a periodic structure that diffracts light into several beams travelling in different directions (i.e., different diffraction angles).
- The emerging coloration is a form of structural coloration.
- Diffraction gratings are commonly used for spectroscopic dispersion and analysis of light.
- They form a sharper pattern than double slits do, with their bright fringes being narrower and brighter while their dark regions are darker.
- A diffraction grating can be chosen to specifically analyze a wavelength emitted by molecules in diseased cells in a biopsy sample or to help excite strategic molecules in the sample with a selected wavelength of light.
- When light is incident on the grating, the split light will have maxima at an angle .
- If a diffraction grating has a grating density of 100 slits per cm, then the slits must be separated by d = 1/100 cm = 10^−4 m. This number can then be used in calculations for the angle at which bright fringes are seen.
- Diffraction is the spreading of a wave due to it going through a gap (Aperture) or around a small object (such as in single slit diffraction).

Spectra with Grating

A diffraction grating is an optical component with a periodic structure that splits and diffracts light into several beams traveling in different directions. The directions of these beams depend on the spacing of the grating and the wavelength of the light so that the grating acts as the dispersive element. Because of this, gratings are commonly used in monochromators and spectrometers.

- The diffraction pattern formed with a grating is known as a grating spectrum .
- The diffraction grating spectrometer has a superiority S= tani, where is the angular length of the slit and i is the angle of incidence onto the grating .
- Gratings form a sharper pattern than double slits do. Their bright fringes are narrower and brighter while their dark regions are darker .
- When very many slits are used in a device called a diffraction grating, the result is very sharp, very bright lines at the points of maximum constructive interference, and darkness everywhere else .

Dispersive Power

Dispersive power is a measure of the ability of a material to separate light of different wavelengths. It is defined as the ratio of the difference in the refractive indices of two wavelengths to the average refractive index of the material for those two wavelengths.

Mathematically, the dispersive power ( ) of a material is given by the formula:

 = (n2 - n1) / (n2 + n1) / 2

where n1 and n2 are the refractive indices of the material for two different wavelengths of light.

The dispersive power of a material is an important property in the field of optics, as it determines the degree to which a prism or lens made of that material will disperse light into its constituent colors. Materials with a high dispersive power will produce a greater separation of colors, while materials with a low dispersive power will produce a smaller separation of colors.

In the context of wave optics, the dispersive power of a material is related to its chromatic aberration, which is the failure of a lens to focus all colors of light to the same point. Materials with a high dispersive power will produce greater chromatic aberration, while materials with a low dispersive power will produce less chromatic aberration.

Overall, the dispersive power of a material is an important property to consider when designing optical systems, as it can affect the quality of the images produced by those systems. It is also an important property to consider when studying the behavior of light as it passes through different materials.

Resolving Power

Resolving power is the ability of an optical instrument to distinguish between two closely spaced objects. It is a measure of the clarity of an image produced by the instrument.

- The resolving power of an instrument is determined by the wavelength of the light used and the numerical aperture of the instrument.
- The Rayleigh criterion states that two point sources are just resolved when the central maximum of one diffraction pattern coincides with the first minimum of the other diffraction pattern.
- The resolving power of a microscope is given by the formula R = 0.61 /NA, where is the wavelength of the light used and NA is the numerical aperture of the objective lens.
- The resolving power of a telescope is given by the formula R = 1.22 /D, where is the wavelength of the light used and D is the diameter of the objective lens or mirror.
- The resolving power of a grating is given by the formula R = nN, where n is the order of the spectrum and N is the total number of lines on the

grating.

- The resolving power of a spectrometer is given by the formula R = /Δ , where is the wavelength of the light used and Δ is the smallest difference in wavelength that can be resolved by the instrument.

Rayleigh's criterion of resolution

Rayleigh's criterion of resolution is a standard for the minimum separation between two light sources that can be resolved by an optical instrument. It is based on the diffraction pattern produced by the instrument's aperture.

- According to Rayleigh's criterion, two point sources are considered to be just resolved when the central maximum of the diffraction pattern of one source coincides with the first minimum of the diffraction pattern of the other source.
- The minimum angular separation between two point sources that can be resolved by an optical instrument is given by the formula = 1.22 /D, where is the angular separation, is the wavelength of the light, and D is the diameter of the instrument's aperture.
- Rayleigh's criterion is commonly used to determine the resolving power of optical instruments such as telescopes and microscopes.
- The criterion is named after the British scientist Lord Rayleigh, who first derived it in the 19th century.

Resolving Power of Grating

The resolving power of a grating is defined as its ability to separate two closely spaced wavelengths of light. It is given by the formula:

```
Resolving Power = Nm
```

Where `N` is the total number of lines on the grating and `m` is the order of the spectrum.

- The resolving power of a grating increases with the increase in the number of lines on the grating and the order of the spectrum.
- The resolving power of a grating is directly proportional to the width of the grating.
- The resolving power of a grating is also dependent on the angle of incidence of light on the grating.
- The resolving power of a grating can be improved by increasing the number of lines on the grating, increasing the width of the grating, or by using light of a smaller wavelength.

Unit 4 - Fiber Optics & Laser

Fiber optics is a technology that uses thin, flexible fibers of glass or other transparent solids to transmit light signals, primarily for telecommunications. The main advantages of fiber optics over traditional copper wire communications include:

1. Higher bandwidth, allowing for faster data transmission.
2. Immunity to electromagnetic interference.
3. Lower signal loss over long distances.
4. Smaller size and weight.

A laser is a device that emits light through a process of optical amplification based on the stimulated emission of electromagnetic radiation. Lasers have many applications, including:

1. Medicine, for example in laser eye surgery.
2. Industry, for example in cutting and welding.
3. Telecommunications, for example in fiber optic communications.
4. Military, for example in target designation and ranging.

Fiber optic communications often use lasers to transmit signals over long distances. The laser light is modulated with the information to be transmitted and then coupled into the fiber optic cable. At the receiving end, the light is detected and the information is extracted from the modulated signal.

Fibre Optics: Principle and construction of optical fiber

Fiber optics is a technology that uses glass or plastic threads (fibers) to transmit data. A fiber optic cable consists of a bundle of glass threads, each of which is capable of transmitting messages modulated onto light waves.

Principle of Fiber Optics

The principle behind fiber optics is based on the transmission of data through light. Light is guided through the core of the fiber optic cable through a process called total internal reflection. This occurs when light is transmitted at an angle greater than the critical angle, causing it to be reflected back into the core of the fiber.

Construction of Optical Fiber

An optical fiber is made up of three parts: the core, the cladding, and the buffer coating. The core is the central part of the fiber, where the light is transmitted. It is made of glass or plastic and has a high refractive index. The cladding surrounds the core and has a lower refractive index, which causes the light to be reflected back into the core. The buffer coating is a protective layer that surrounds the cladding and provides mechanical protection to the fiber.

The process of making an optical fiber involves drawing glass or plastic to a diameter slightly thicker than that of a human hair. The core and cladding are made from different materials, with the core having a higher refractive index.

The materials are heated and drawn together to form a preform, which is then drawn to the desired diameter.

In summary, fiber optics is a technology that uses light to transmit data through glass or plastic fibers. The principle behind it is total internal reflection, and the construction of an optical fiber involves the core, cladding, and buffer coating. The process of making an optical fiber involves drawing glass or plastic to a diameter slightly thicker than that of a human hair.

Acceptance Angle

- The acceptance angle of an optical fiber is defined based on a purely geometrical consideration (ray optics).
- It is the maximum angle of a ray (against the fiber axis) hitting the fiber core which allows the incident light to be guided by the core.
- Acceptance angle is just the conical half angle of the acceptance cone.
- Acceptance angle is the maximum angle with the axis of the optical fiber at which light can enter the fiber, in order to be propagated through it.

Numerical Aperture

Numerical aperture (NA) is a measure of the light-gathering ability of an optical system. It is commonly used in microscopy to describe the acceptance cone of an objective, and in fiber optics, it describes the range of angles within which light that is incident on the fiber will be transmitted along it .

In fiber optics, the numerical aperture is considered as the light-gathering capacity of an optical fiber. It is defined as the sine of half of the angle of the fiber's light acceptance cone, i.e., NA = sin(a), where a is called the acceptance cone angle .

In laser physics, the definition of numerical aperture is slightly different. Laser beams spread out as they propagate, but slowly. Far away from the narrowest part of the beam, the spread is roughly linear with distance, forming a cone of light in the "far field" .

Acceptance cone for the notes of the Unit 4 - Fiber Optics & Laser in the subject of ENGINEERING PHYSICS

- The acceptance cone is the cone within which optical power may be coupled into the bound modes of an optical fiber .
- The cross section of an optical fiber is circular; the light waves accepted by the core are expressed as a cone .
- The larger the acceptance cone, the larger the numerical aperture (NA); this means that the fiber is able to accept and propagate more light .
- The acceptance cone is derived by rotating the acceptance angle about the fiber axis .

- In order to get high light transmission, light should propagate through the fiber by a series of total internal reflections. This transmission is achieved if the angles of the light entering the fiber are within the acceptance cone .

Unit 4 - Fiber Optics & Laser: Step Index and Graded Index Fibers

1. **Step Index Fiber:** A step index fiber is a type of optical fiber that has a uniform refractive index within the core and a sharp decrease in refractive index at the core-cladding interface. This results in a step-like change in the refractive index profile.

2. **Graded Index Fiber:** A graded index fiber, on the other hand, has a varying refractive index within the core. The refractive index gradually decreases from the center of the core to the core-cladding interface. This results in a smooth change in the refractive index profile.

3. **Difference between Step Index and Graded Index Fiber:** The main difference between step index and graded index fibers is the way light travels through the fiber. In a step index fiber, light travels in a straight line until it reaches the core-cladding interface, where it is reflected back into the core. In a graded index fiber, light travels in a curved path due to the varying refractive index within the core.

4. **Advantages of Graded Index Fiber:** Graded index fibers have several advantages over step index fibers. They have a higher bandwidth, which means they can transmit more data at a faster rate. They also have lower attenuation, which means the signal is less likely to degrade over long distances.

5. **Applications of Step Index and Graded Index Fibers:** Both step index and graded index fibers are used in a variety of applications, including telecommunications, data transmission, and sensing. Graded index fibers are often used in high-speed data transmission, while step index fibers are commonly used in sensing applications.

Fiber Optic Communication Principle

Fiber optic communication is a method of transmitting information from one place to another by sending pulses of infrared light through an optical fiber. The light is a form of carrier wave that is modulated to carry information.

The fiber optic works on the principle of total internal reflection. When a ray of light is incident on the core of the optical fiber at a small angle, it suffers refraction and strikes the core cladding interface. As the diameter of the fiber is very small, the angle of incidence is greater than the critical angle.

Optical fiber is used by telecommunications companies to transmit telephone signals, Internet communication, and cable television signals. It is also used in other industries, including medical, defense, government, industrial, and commercial.

First, an electrical signal is converted into an optical signal, and then an optical signal is transmitted through the optical fiber, which is a type of wired communication.

In telecommunications, fiber optic technology has virtually replaced copper wire in long-distance telephone lines, and it is used to link computers within local area networks.

Attenuation in Fiber Optics & Laser

Attenuation in fiber optics, also known as transmission loss, is the reduction in intensity of the light beam (or signal) with respect to distance traveled through a transmission medium. Attenuation coefficients in fiber optics usually use units of dB/km through the medium due to the relatively high quality of transparency of modern optical transmission media.

The attenuation coefficient ( ) of the optical fiber can be determined for specific wavelengths. For example, it has been measured for wavelengths of 532 nm and 671 nm. The attenuation coefficient ( ) is measured as a function of the laser input power (Pin) and the optical fiber length. The results show that the power attenuation of the laser beam is wavelength dependent.

Fiber optic systems transmit in the "windows" created between the absorption bands at 850 nm, 1300 nm, and 1550 nm, where physics also allows one to fabricate lasers and detectors easily.

Attenuation (dB) can be calculated using the formula: Attenuation (dB) = 10 x Log 10 (P in/P out) = 20xLog 10 (V in/V out). Gain (dB) can be calculated using the formula: Gain (dB) = 10 x Log 10 (P out/P in) = 20 x Log 10 (V out/V in).

Fiber attenuation and dispersion are the major limiting factors in an optical network. These fiber attributes define the spans between regenerators and the signal transmission rates. Recent advances in fiber designs and processing technologies have been made to overcome these limiting factors.

Dispersion in Fiber Optics

Dispersion is a property of telecommunication signals along transmission lines, such as the pulses of light in optical fiber . It is the spreading of light pulses when the wave travels through an optical fiber from one end to another . This results in some variation of the actually transmitted wave through an optical fiber . Due to the dispersion of light waves, various adverse effects are noticed on the signal being transmitted .

There are two main types of dispersion in fiber optics: material dispersion and waveguide dispersion . Material dispersion is caused by a change in the fiber optic material's refractive index with different wavelengths . The higher the index, the slower the light travels . Waveguide dispersion is due to the distribution of light between the core and cladding .

Dispersion distorts signals and limits the data rate of digital signals sent over fiber optic cable . In this section, we analyze this dispersion and its effect on digital signals . Figure 8.3.1 shows the variety of paths that light may take through a straight fiber optic cable .

Unit 4 - Fiber Optics & Laser: Application of Fiber

1. **Telecommunications**: Optical fibers are widely used in telecommunications to transmit data over long distances. They have a higher bandwidth than traditional copper cables, allowing for faster data transfer.
2. **Medical**: Optical fibers are used in medical procedures such as endoscopy, where they are used to transmit light into the body to allow doctors to see inside.
3. **Sensors**: Optical fibers can be used as sensors to detect changes in temperature, pressure, and strain. They are used in structural health monitoring, oil and gas exploration, and other applications.
4. **Lighting**: Optical fibers are used in lighting to transmit light from a source to a remote location. They are used in decorative lighting, signage, and other applications.
5. **Military**: Optical fibers are used in military applications for secure communication, as they are difficult to tap into without detection.
6. **Industrial**: Optical fibers are used in industrial applications for sensing and control. They are used in manufacturing, robotics, and other applications.

Laser: Absorption of radiation

1. Absorption of radiation is the process by which an atom or molecule takes up energy from electromagnetic radiation.
2. When an atom or molecule absorbs a photon, one of its electrons is excited to a higher energy level.
3. The energy of the absorbed photon must match the energy difference between the initial and final energy levels of the electron.
4. The probability of absorption depends on the intensity of the radiation and the absorption cross-section of the atom or molecule.
5. Absorption can be used to measure the concentration of a substance in a sample by measuring the amount of radiation absorbed by the sample.
6. In lasers, absorption of pump radiation by the laser medium is a crucial step in the process of generating laser light.
7. The absorption of pump radiation excites the electrons in the laser medium to higher energy levels, creating a population inversion necessary for laser action.

8. The efficiency of absorption of pump radiation is an important factor in the design of a laser, as it affects the overall efficiency of the laser.

Spontaneous and Stimulated Emission of Radiation

Spontaneous and stimulated emission of radiation are two processes that are fundamental to the operation of lasers. These processes are related to the interaction of light with matter, specifically with the electrons in atoms or molecules.

1. **Spontaneous emission** is the process by which an excited atom or molecule, with an electron in a higher energy level, releases a photon and returns to a lower energy level. This process is random and does not require any external influence.

2. **Stimulated emission** is the process by which an incoming photon, with the same energy as the energy difference between two levels of an atom or molecule, causes the atom or molecule to release a second photon and return to a lower energy level. This process is not random and requires the presence of an incoming photon with the right energy.

The key difference between spontaneous and stimulated emission is that spontaneous emission is a random process, while stimulated emission is a triggered process. In a laser, stimulated emission is used to amplify light, while spontaneous emission is a source of noise that can limit the performance of the laser.

Population Inversion

- Population inversion is a technique mainly used for light amplification .
- It is required for laser operation .
- Consider a group of electrons with two energy levels E1 and E2 .
- A population inversion (N2 > N1) has thus been achieved between level 1 and 2, and optical amplification at the frequency 21 can be obtained .
- At least half the population of atoms must be excited from the ground state to obtain a population inversion, the laser medium must be very strongly pumped .
- Population inversion, and thus optical gain in the sense of Eq. 14, may be obtained in a direct bandgap semiconductor exhibiting radiative recombination of electrons and holes provided there is, simultaneously in space and time, a large density of electrons in the conduction band and holes in the valence band such as that provided by the junction region of a diode under heavy forward bias where large numbers of electrons and holes exist .
- Population inversion and laser action have been discovered in a large number of optical transitions in gases and gas mixtures .
- A gas as the laser active medium has several advantages: scalability to large volume and effective cooling by diffusion or by convective flow, allowing potentially high power output .

- Normally, a system of atoms is in temperature equilibrium and there are always more atoms in low energy states than in higher ones .

Einstein's Coefficients

Einstein's coefficients are mathematical quantities that measure the probability of absorption or emission of light by an atom or molecule. These coefficients are determined by the atomic system and are fixed probabilities per time associated with each atom. They do not depend on the state of the gas of which the atoms are a part. Therefore, any relationship that we can derive between the coefficients at, say, thermodynamic equilibrium will be valid universally.

There are three Einstein coefficients: A21, B12, and B21. The Einstein A coefficients are related to the rate of spontaneous emission of light, while the Einstein B coefficients are related to the absorption and stimulated emission of light .

Einstein's theory of absorption and emission of light by an atom is based on Planck's theory of radiation. Under thermal equilibrium, the population of energy levels obeys the Boltzmann's distribution law.

The ratio of Einstein's coefficient of spontaneous emission to the Einstein's coefficient of stimulated absorption is proportional to the cube of frequency v.

Principles of Laser Action

1. **Population Inversion:** In order to achieve laser action, the number of atoms in the excited state must be greater than the number of atoms in the ground state. This condition is known as population inversion.
2. **Optical Feedback:** The laser cavity must be designed in such a way that the light emitted by the excited atoms is reflected back into the cavity, stimulating further emission of light. This is achieved by using a pair of mirrors, one of which is partially transparent, to form an optical resonator.
3. **Stimulated Emission:** When an atom in the excited state is struck by a photon of the same energy as the energy difference between the excited state and the ground state, the atom can be stimulated to emit a second photon of the same energy, in phase with the first photon. This process is known as stimulated emission.
4. **Gain Medium:** The gain medium is the material in which the population inversion is achieved. It can be a gas, liquid, or solid, and must have the appropriate energy levels to support laser action.
5. **Pumping:** In order to achieve population inversion, the atoms in the gain medium must be excited to the higher energy state. This is achieved by pumping energy into the gain medium, using an external energy source such as an electrical discharge or a flashlamp.

These are the basic principles of laser action that are essential for the operation of a laser. They are important concepts to understand when studying the subject of Engineering Physics, specifically in the unit on Fiber Optics and Lasers.

Unit 4 - Fiber Optics & Laser in the subject of ENGINEERING PHYSICS

Solid state Laser (Ruby laser) and Gas Laser (He-Ne laser)

- **Ruby Laser**: A ruby laser is a three-level solid-state laser. In a ruby laser, an optical pumping technique is used to supply energy to the laser medium. Optical pumping is a technique in which light is used as an energy source to raise electrons from a lower energy level to a higher energy level .
- **Solid-state lasers**: Solid-state lasers include all optically pumped lasers in which the gain medium is a solid at room temperature. In solid-state laser materials, the atoms responsible for generating laser light are first excited to higher energy states through the absorption of photons. The way in which these atoms relax from their excited states determines the quality and quantity of laser light produced .
- **He-Ne Laser**: The first He-Ne laser, a gas laser, followed in 1961. It is a gas laser built by Ali Javan at MIT, with a wavelength of 632.8 nm and a linewidth of only 10kHz .
- **Solid-state lasers vs Gas lasers**: Solid-state lasers utilize crystals or glass materials doped with transition metal or rare earth ions. They are often utilized for high power applications, such as material processing and medical lasers, because they can reach some of the highest peak powers out of all commercially available lasers. However, it is harder to cool solid-state lasers compared to gas lasers .

Laser Applications in Fiber Optics and Engineering Physics

Lasers have a wide array of applications in the field of Fiber Optics and Engineering Physics. Some of the applications are:

1. **Lasers in Communication:** Fiber optic cables are a major mode of communication partly because multiple signals can be sent with high quality and low loss by light propagating along the fibers. The light signals can be modulated with the information to be sent by either light emitting diodes or lasers.
2. **Fiber Lasers:** Fiber lasers use an optical fiber cable made of silica glass to guide light. The resulting laser beam is more precise than with other types of lasers because it is straighter and smaller. They also have a small footprint, good electrical efficiency, low maintenance, and low operating costs.

3. **Optical Fiber Communications Systems:** Optical fiber communications systems today commonly use Erbium-doped fiber amplifiers (EDFA's) that amplify ~1.5-micron wavelength signals having bandwidths up to ~4 THz.

4. **Fiberoptics Applications in Sensors and Communications:** Fiberoptics have applications in sensors and communications. Future developments in this field are expected.

5. **Other Applications:** Lasers are essential to an incredibly large number of applications. Today, they are used in bar code readers, compact discs, medicine, communications, sensors, materials processing, computer printers, data processing, 3D-imaging, spectroscopy, navigation, non-destructive testing, chemical processing, color copiers, etc.

Unit 5 - Superconductors and Nano-Materials:

Superconductors are materials that have zero electrical resistance and can conduct electricity with 100% efficiency. This means that when a current is applied to a superconductor, it will continue to flow indefinitely without any energy loss. Superconductivity is a phenomenon that occurs at very low temperatures, typically below -200°C.

Nano-materials, on the other hand, are materials that have at least one dimension in the nanometer range (1-100 nm). These materials have unique properties due to their small size and high surface area to volume ratio. Some common examples of nano-materials include carbon nanotubes, graphene, and quantum dots.

Some key points to remember about superconductors and nano-materials are:

1. Superconductors have zero electrical resistance and can conduct electricity with 100% efficiency.
2. Superconductivity occurs at very low temperatures, typically below -200°C.
3. Nano-materials have at least one dimension in the nanometer range (1-100 nm).
4. Nano-materials have unique properties due to their small size and high surface area to volume ratio.
5. Common examples of nano-materials include carbon nanotubes, graphene, and quantum dots.

Superconductors: Temperature dependence of resistivity in superconducting materials

- Superconductors are special materials in which after a certain critical temperature the resistance drops to zero value and the conductivity thus reaches the maximum .

- The critical temperature is known as TC and is the temperature below which the resistivity of the material drops to zero .
- These temperatures are low, with most industrial superconductors below 12 K and higher temperature ceramics just above 130K .
- The superconductivity of material is influenced by external temperature, current, and external magnetic field .
- The resistivity of materials depend on the temperature. The equation that shows the relation between the temperature and the resistivity of a material is: t = 0 [1 + (T − T 0)] .
- In the equation, 0 is the resistivity at a standard temperature, t is the resistivity at t 0 C, T 0 is the reference temperature and is the temperature coefficient of resistance .

Meissner effect for the notes of the Unit 5 - Superconductors and Nano-Materials: in the subject of ENGINEERING PHYSICS

- The Meissner effect is the expulsion of a magnetic field from the interior of a material that is in the process of becoming a superconductor.
- This occurs when the material is cooled below a certain temperature, called the transition temperature, usually close to absolute zero.
- When a magnetic field is applied at a higher value of temperature to the transition temperature, the superconductor has a minor effect.
- One phenomena that occurs in superconductors below the critical temperature is the Meissner effect, which is where a superconductor expels all magnetic field from within itself.
- One of the most well-known demonstrations of the Meissner effect is its ability to make a magnet levitate above a superconductor.
- All superconductors, whether `conventional` or `unconventional`, exhibit the Meissner effect.

Temperature dependence of critical field

The critical field of a superconductor is the magnetic field strength at which the superconductor transitions from the superconducting state to the normal state. The critical field is dependent on the temperature of the superconductor.

- At absolute zero temperature, the critical field is at its maximum value, known as the thermodynamic critical field.
- As the temperature of the superconductor increases, the critical field decreases.
- The relationship between the critical field and temperature can be described by the empirical formula Hc(T) = Hc(0)(1-(T/Tc)^2), where Hc(0) is the critical field at absolute zero temperature, T is the temperature of the superconductor, and Tc is the critical temperature of the superconductor.
- The critical field also depends on the material properties of the superconductor, such as its purity and crystal structure.

- The temperature dependence of the critical field is an important factor in the design and operation of superconducting devices, as it determines the operating temperature range and magnetic field strength that the device can withstand.

Persistent Current

- Persistent current refers to a perpetual electric current that does not require an external power source.
- Such a current is impossible in normal electrical devices, since all commonly-used conductors have a non-zero resistance, and this resistance would rapidly dissipate any such current as heat.
- However, in superconductors and some mesoscopic devices, persistent currents are possible and observed due to quantum effects.
- In resistive materials, persistent currents can appear in microscopic samples due to size effects.
- Persistent currents are widely used in the form of superconducting magnets.
- When a voltage is applied to a round superconductor and then disconnected, the current will flow round and round forever. This is called a persistent current.
- This persistent current produces a magnetic flux which makes the magnetic flux passing through the ring constant.

Type I and Type II Superconductors

Type I Superconductors

- Type I superconductors have only one critical magnetic field. When that value is reached, the superconductive characteristics of a material are gone.
- The transition from normal state to superconducting state occurs instantly.
- Magnetic field lines cannot penetrate through Type I superconductors.
- Type I superconductors are generally pure metals.

Type II Superconductors

- Type II superconductors have two critical magnetic fields.
- The transition from normal state to superconducting state happens slowly. By decreasing temperature after critical temperature, superconducting properties increase.
- Some magnetic field lines can penetrate through Type II superconductors.
- Most Type II superconductors have higher critical temperature.
- Type II superconductors are usually made of metal alloys or complex oxide ceramics. All high-temperature superconductors are Type II superconductors. While most elemental superconductors are Type I, niobium,

vanadium, and technetium are elemental Type II superconductors. Boron-doped diamond and silicon are also Type II superconductors.

- Type II superconductors lose their superconductivity gradually but not easily or abruptly when placed in an external magnetic field.

High temperature superconductors

- High-temperature superconductors (abbreviated high-Tc or HTS) are defined as materials that behave as superconductors at temperatures above 77 K (−196.2 °C; −321.1 °F), the boiling point of liquid nitrogen.
- The main class of high-temperature superconductors is copper oxides combined with other metals, especially the Rare-earth barium copper oxides (REBCOs) such as Yttrium barium copper oxide (YBCO).
- The second class of high-temperature superconductors in the practical classification is the iron-based compounds.
- High-temperature superconductors (HTSCs), when discovered in 1986, presented a huge challenge but also opportunity to companies like ABB, GE, and Siemens to radically improve their products.
- A new experiment led by the condensed matter physicist Séamus Davis at the University of Oxford all but settles the origin of high-temperature superconductivity, a puzzle Davis has worked on for 25 years. When electrons couple up, further quantum trickery makes superconductivity unavoidable.

Properties and Applications of Super-conductors

Properties of Superconductors

- A superconductor is an element or metallic alloy which, when cooled below a certain threshold temperature, the material dramatically loses all electrical resistance.
- In principle, superconductors can allow electrical current to flow without any energy loss (although, in practice, an ideal superconductor is very hard to produce).
- Another important property of a superconducting material is its critical magnetic field Bc(T), which is the maximum applied magnetic field at a temperature T that will allow a material to remain superconducting.

Applications of Superconductors

- Superconductors are used in a variety of applications, but most notably within the structure of the Large Hadron Collider. The tunnels that contain the beams of charged particles are surrounded by tubes containing powerful superconductors.
- Superconductors are used in particle accelerators, generators, transportation, computing, electric motors, medical, power transmission, etc.

- Superconductors are primarily employed for creating powerful electromagnets in MRI scanners. These conductors are used to transmit power for long distances.
- Most of the applications of the superconductor examples are because of their properties which provide advantages such as low power loss because of less dissipation of energy, high-speed operations because of zero resistance and continuous flowing electrical current, and high sensitivity.

Nano-Materials: Introduction and properties of nano materials

Nano-materials are materials that have at least one dimension in the nanometer range (1-100 nm). These materials exhibit unique properties due to their small size and high surface area to volume ratio. Some of the properties of nano-materials include:

1. **High surface area:** Nano-materials have a high surface area to volume ratio, which can result in increased reactivity and catalytic activity.
2. **Quantum confinement:** When the size of a material is reduced to the nanoscale, the movement of electrons can be restricted, resulting in quantum confinement. This can lead to changes in the electronic, optical, and magnetic properties of the material.
3. **Size-dependent properties:** The properties of nano-materials can change as their size is reduced. For example, the melting point of a material can decrease as its size is reduced.
4. **Surface Plasmon Resonance:** Nano-materials can exhibit surface plasmon resonance, which is the collective oscillation of electrons on the surface of the material. This can result in enhanced optical properties, such as increased absorption or scattering of light.

These unique properties make nano-materials useful in a wide range of applications, including electronics, medicine, energy, and materials science. In the field of engineering physics, nano-materials are studied for their potential use in the development of new technologies and materials.

Unit 5 - Superconductors and Nano-Materials: Basics concept of Quantum Dots

Quantum dots are colloidal fluorescent semiconductor nanocrystals, roughly spherical and typically have unique optical, electronic and photophysical properties that make them appealing in promising applications in biological labeling, imaging, and detection and as efficient fluorescence resonance energy transfer donors.

- Quantum dots (QDs) are also called semiconductor nanocrystals.
- They are semiconductor particles a few nanometres in size.
- Their optical and electronic properties differ from those of larger particles as a result of quantum mechanics.
- They are a central topic in nanotechnology and materials science.

- Quantum dots defined by lithographically patterned gate electrodes, or by etching on two-dimensional electron gases in semiconductor heterostructures can have lateral dimensions between 20 and 100 nm.
- Some quantum dots are small regions of one material buried in another with a larger band gap.
- Quantum dots are also brighter than a rival technology known as organic LEDs (OLEDs) and could potentially make OLED displays obsolete.

Quantum Wires and Quantum Wells

Quantum wires and quantum wells are semiconductor structures that are used in nanoelectronics, optoelectronics, information technology, and biotechnology. These structures have applications due to the quantum mechanical confinement provided by the small spatial extent of the structures.

A quantum well is a two-dimensional (2D) structure where the particle motion is quantized in one direction, while the particle is free to move in the other two directions. Quantum well materials are made of thin layers of differing semiconductor materials, and can be grown such that the movement of the free electron is confined to a two-dimensional plane.

A quantum wire is a one-dimensional (1D) structure where the particle motion is quantized in two directions, leading to free movement along only one direction. For metals, quantization corresponding to the lowest energy states is only observed for atomic wires. Their corresponding wavelength being thus extremely small they have a very large energy separation which makes resistance quantization observable even at room temperature.

Quantum wells and quantum well devices are a subfield of solid-state physics that is still extensively studied and researched today. The theory used to describe such systems uses important results from the fields of quantum physics, statistical physics, and electrodynamics.

Fabrication of Nano Materials - Top-Down Approach (CVD) and Bottom-Up Approach (Sol Gel)

Unit 5 - Superconductors and Nano-Materials: Engineering Physics

Nano materials are materials with structures at the nanoscale, typically between 1 and 100 nanometers. There are two main approaches to fabricating nano materials: the top-down approach and the bottom-up approach.

Top-Down Approach (CVD)

The top-down approach involves breaking down larger materials into smaller nano-sized particles. One common method used in the top-down approach is Chemical Vapor Deposition (CVD). In CVD, a gas-phase precursor is introduced into a reaction chamber, where it decomposes and deposits onto a substrate to form a thin film of the desired material. The film can then be patterned and etched to create nanostructures.

Bottom-Up Approach (Sol Gel)

The bottom-up approach, on the other hand, involves building up nanostructures from smaller building blocks, such as atoms or molecules. One common method used in the bottom-up approach is the Sol Gel process. In the Sol Gel process, a solution of precursor molecules is prepared and then allowed to undergo a controlled chemical reaction to form a gel. The gel is then dried and heat-treated to form a solid material with nano-sized pores and particles.

Both the top-down and bottom-up approaches have their advantages and disadvantages, and the choice of approach depends on the specific application and desired properties of the nano material. For example, the top-down approach is often used for creating nanostructures with well-defined shapes and sizes, while the bottom-up approach is often used for creating materials with high surface area and porosity.

Properties and Application of Nano Materials

Nanomaterials are materials with a particle size of less than 100 nm in at least one of its dimensions. They possess unique optical, magnetic, electrical, and other properties that emerge at this scale. These emergent properties have great potential applications in various fields, including electronics, medicine, and biomedicine .

Some of the properties and applications of nanomaterials are:

1. **Optical properties**: The optical properties of nanomaterials are used to form optical detectors, sensors, lasers, displays, and solar cells. This property is also used in biomedicine and photoelectrochemistry.
2. **Electrical properties**: Nanomaterials can be used in microbial fuel cells, where the electrodes are made up of carbon nanotubes.
3. **Medicine**: Nanoparticles may find practical applications in medicine.
4. **Engineering**: Nanoparticles may find practical applications in engineering.
5. **Catalysis**: Nanoparticles may find practical applications in catalysis.
6. **Environmental remediation**: Nanoparticles may find practical applications in environmental remediation.